

Stephen Beckett, Ph.D.

Associate Research Scientist

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&

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updated: May 7, 2026

Areas of focus: marine microbial ecology, virus-microbe ecology and evolution, biophysics, epidemic dynamics, network structure and dynamics of complex systems.

Research skills: mathematical modelling and data science, network analysis, spatial dynamics & time series analysis, data visualization & mapping, development of interactive data dashboards.

Professional Preparation

- 2024-Present **Associate Research Scientist**
University of Maryland Institute for Health Computing
- 2023-Present **Associate Research Scientist**
Department of Biology, University of Maryland
- 2023- 2024 **Affiliate**
School of Biological Sciences, Georgia Institute of Technology
- 2022-2023 **Research Scientist II**
School of Biological Sciences, Georgia Institute of Technology
- 2019-2022 **Research Scientist I**
School of Biological Sciences, Georgia Institute of Technology
- 2015-2019 **Postdoctoral Fellow**
School of Biological Sciences, Georgia Institute of Technology
- 2011-2015 **Ph.D. Biological Sciences**
Biosciences, University of Exeter
Thesis: Nestedness and modularity in bipartite networks. [info:hdl/10871/17913](https://hdl.handle.net/10871/17913)
- 2011 **Research Intern**
Computational Ecology and Environmental Science, Microsoft Research Cambridge.
- 2010-2011 **MRes Mathematics in the Living Environment**
University of York
- 2007-2010 **BSc (Hons) Geography and Mathematics**
University of Leeds

Honours and Awards

- 2018 Climate Change Fellow at *Georgia Institute of Technology*.
- 2013 Travel grant from the *Society of General Microbiology* to attend Autumn conference.
- 2011-2015 Research & Knowledge Transfer PhD studentship from the *University of Exeter*.
- 2010-2011 Partially funded masters studentship from *Natural Environment Research Council*.

Editorial Boards

2025-Present *Academic Editor*, PLOS Computational Biology

2024-Present *Associate Editor*, Mathematics in Medical and Life Sciences

Funding

- 7/1/2024-6/30/2025 UMD Pandemic Readiness Initiative: Towards a wastewater surveillance and communications nexus (**Beckett, PI** & McPhaul Co-PI, Weitz Co-PI, \$48,400).
- 9/21/2022-9/20/2023 Centers for Disease Control and Prevention: Mathematical Modeling – COVID-19 Mathematical Modeling of Healthcare Impact and Capacity- Georgia Institute of Technology (Weitz, PI & **Beckett, Aim Lead**, \$300,000). [note, Beckett took over as PI from July-September 2023].
- 10/1/2021-3/15/2022 Rockefeller Foundation Covid-19 Event Risk Assessment Deployment (Weitz, PI & **Beckett Co-PI**, \$50,371)
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Publications (* = Joint lead authors, † = undergraduate research mentee)

1. Frémont P., **Beckett S.J.**, Demory D., Carr E., Follett C.F., Lindell D., Talmy D., Dutkiewicz S., Weitz J.S. Coexistence of Photosynthetic Marine Microorganisms, Viruses, and Grazers: Toward Integration in Ocean Ecosystem Models. *Environmental Microbiology* 28: e70295. DOI: [10.1111/1462-2920.70295](https://doi.org/10.1111/1462-2920.70295).
Associated code: https://github.com/PaulFremont3/SIVZ_coexistence.
2. Dey, R., Coenen, A.R., Solonenko, N., Burris M.N., Mackey A.I., Galasso J., Sun C.L., Demory D., Muratore D., **Beckett S.J.**, Sullivan, M.B., Weitz J.S. Density-dependent inhibition and higher-order interactions enable coexistence in phage-bacteria community dynamics. *The ISME Journal* 20: wrag041. DOI: [10.1093/ismejo/wrag041](https://doi.org/10.1093/ismejo/wrag041).
Associated code: https://github.com/RaunakDey/VIMIMO_paper.
3. Cornfeld, O., Niu, K., Neeman O., Roswell, M., Steinbach G., **Beckett S.J.**, Wardi Y., Weitz J.S., Yashiv, E. (2025) Integrating macroeconomic and public health impacts in social planning policies for pandemic response. *Epidemics* 53: 100873. DOI: [10.1016/j.epidem.2025.100873](https://doi.org/10.1016/j.epidem.2025.100873).
Associated code: <https://github.com/odedneeman/Optimal-Pandemic-Control>.
4. Harris J.D., Gallmeier E., Dushoff J., **Beckett S.J.**, Weitz J.S. (2025) Infections are not alike: the effects of covariation between individual susceptibility and transmissibility on epidemic dynamics. *Journal of Theoretical Biology* 613: 112211. DOI: [10.1016/j.jtbi.2025.112211](https://doi.org/10.1016/j.jtbi.2025.112211).
Associated code: https://github.com/Jeremy-D-Harris/SIR_heterogeneity_project.
5. Goel T., **Beckett S.J.**, Weitz J.S. (2025) Eco-evolutionary dynamics of temperate phages in periodic environments. *Virus Evolution* 11(1): veaf019. DOI: [10.1093/ve/veaf019](https://doi.org/10.1093/ve/veaf019).
Associated code: <https://github.com/tapangoel1994/EcoEvoDynamicsInPeriodicEnvironments>.
6. **Beckett S.J.***, Demory D.*, Coenen A.R., Casey J.R., Dugenne M., Follett C.L., Connell P., Carlson M.C.G., Hu S.K., Wilson S.T., Muratore D., Rodriguez-Gonzalez R.A., Peng S., Becker K.W., Mende D.R., Armbrust E.V., Caron D.A., Lindell D., White A.E., Ribalet F., Weitz J.S. (2024) Disentangling top-down drivers of mortality underlying diel population dynamics of *Prochlorococcus* in the North Pacific Subtropical Gyre. *Nature Communications* 15: 2105. DOI: [10.1038/s41467-024-46165-3](https://doi.org/10.1038/s41467-024-46165-3).
Associated material: <https://doi.org/10.5281/zenodo.7388552>.

7. Sinclair A.H., Taylor M.K., Davidson A., Weitz J.S., **Beckett S.J.**, Samanez-Larkin G.R. (2024) Scenario-Based Messages on Social Media Motivate COVID-19 Information Seeking. *Journal of Applied Research in Memory and Cognition* 13(1): 124–135. DOI: [10.1037/mac0000114](https://doi.org/10.1037/mac0000114). Associated material: <https://doi.org/10.17605/OSF.IO/MBH9W>.
8. Sinclair A.H., Taylor M.K., Brandel-Tanis F., Davidson A., Chande A.T., Rishishwar L., Andris C., Adcock R. A., Weitz J.S., Samanez-Larkin G.R., **Beckett S.J.** (2023) Communicating COVID-19 exposure risk with an interactive website counteracts risk misestimation. *PLoS ONE* 18(10): e0290708. DOI: [10.1371/journal.pone.0290708](https://doi.org/10.1371/journal.pone.0290708). Associated material: <https://doi.org/10.17605/OSF.IO/MBH9W>.
9. **Beckett S.J.**, Brandel-Tanis F.A., Nguyen Q.†, Chande A.T., Rishishwar L., Andris C., Weitz J.S. (2023) localcovid19now: processing and mapping COVID-19 case data at subnational scales. *Journal of Open Source Software* 8(81): 4898. DOI: [10.21105/joss.04898](https://doi.org/10.21105/joss.04898). Associated code: <https://github.com/sjbeckett/localcovid19now>.
10. Sinclair A.H., Taylor M.K., Weitz J.S., **Beckett S.J.**, Samanez-Larkin G.R. (2023) Reasons for Receiving or Not Receiving Bivalent COVID-19 Booster Vaccinations Among Adults — United States, November 1–December 10, 2022. *MMWR Morbidity and Mortality Weekly Report* 72(3): 73–75. DOI: [10.15585/mmwr.mm7203a5](https://doi.org/10.15585/mmwr.mm7203a5).
11. Muratore D., Boysen A.K., Harke M.J., Becker K.W., Casey J.R., Coesel S.N., Mende D.R., Wilson S.T., Aylward F.O., Eppley J.M., Vislova A., Peng S., Rodriguez-Gonzalez R.A., **Beckett S.J.**, Armbrust E.V., DeLong E.F., Karl D.M., White A.E., Zehr J.P., Van Mooy B.A.S., Dyhrman S.T., Ingalls A.E., Weitz J.S. (2022) Complex Marine Microbial Communities Partition Metabolism of Scarce Resources Over the Diel Cycle. *Nature Ecology & Evolution* 6: 218–229. DOI: [10.1038/s41559-021-01606-w](https://doi.org/10.1038/s41559-021-01606-w).
12. Gibson G., Weitz J.S., Shannon M.P., Holton B., Bryskin A., Liu B., Sieglinger M., Coenen A.R., Zhao C., **Beckett S.J.**, Bramblett S., Williamson J., Farrell M., Ortiz A., Abdallah C.T., García A.J. (2022) Surveillance-to-Diagnostic Testing Program for Asymptomatic SARS-CoV-2 Infections on a Large, Urban Campus in Fall 2020. *Epidemiology* 33: 209–216. DOI: [10.1097/EDE.0000000000001448](https://doi.org/10.1097/EDE.0000000000001448). Associated code: https://github.com/jsweitz/gtcovid_fa20_spr21_analytics.
13. Lopman B.A., Shioda K., Nguyen Q.†, **Beckett S.J.**, Siegler A.J., Sullivan P.S., Weitz J.S. (2021) A framework for monitoring population immunity to SARS-CoV-2. *Annals of Epidemiology* 63: 75–78. DOI: [10.1016/j.annepidem.2021.08.013](https://doi.org/10.1016/j.annepidem.2021.08.013). Associated code: <https://github.com/quannnguyenminh103/Covid19-Population-Level-Immunity>. Associated website: <https://popimmunity.biosci.gatech.edu/>.
14. Mruwat N., Carlson M.C.G., Goldin S., Ribalet F., Kirzner S., Hulata Y., **Beckett S.J.**, Shitrit D., Weitz J.S., Armbrust E.V., Lindell D. (2021) A single-cell polony method reveals low levels of infected *Prochlorococcus* in oligotrophic waters despite high cyanophage abundances. *The ISME Journal* 15: 41–54. DOI: [10.1038/s41396-020-00752-6](https://doi.org/10.1038/s41396-020-00752-6).
15. Chande A., Lee S., Harris M., Nguyen Q.†, **Beckett S.J.**, Hilley T., Andris C., Weitz J.S. (2020) Real-time, interactive website for US-county-level COVID-19 event risk assessment. *Nature Human Behaviour* 4: 1313–1319. DOI: [10.1038/s41562-020-01000-9](https://doi.org/10.1038/s41562-020-01000-9). Associated code: <https://github.com/appliedbinf/covid19-event-risk-planner>. Associated website: <https://covid19risk.biosci.gatech.edu/>.
16. Weitz J.S., **Beckett S.J.**, Coenen A.R., Demory D., Dominguez-Mirazo M., Dushoff J., Leung C-Y., Li G., Măgălie A., Park S.W., Rodriguez-Gonzalez R.A., Shivam S., Zhao C.Y. (2020) Modeling Shield Immunity to Reduce COVID-19 Epidemic Spread. *Nature Medicine* 26: 849–854. DOI:

[10.1038/s41591-020-0895-3](https://doi.org/10.1038/s41591-020-0895-3).

Associated code: https://github.com/WeitzGroup/covid_shield_immunity.

17. Benedetto B., Coenen A.R., **Beckett S.J.**, McGillicuddy Jr. D.J., Weitz J.S., Karl D.M. (2019) The ecological and biogeochemical state of the North Pacific Subtropical Gyre is linked to sea surface height. *Journal of Marine Research* 77: 215-245. DOI: [10.1357/002224019828474241](https://doi.org/10.1357/002224019828474241).
Associated code: <https://github.com/duebi/mesoHot>.
18. Talmy D., **Beckett S.J.**, Taniguchi D.A.A., Brussaard C.P.D., Weitz J.S., Follows, M.J. (2019) An empirical model of carbon flow through marine viruses and microzooplankton grazers. *Environmental Microbiology* 21(6): 2171-2181. DOI: [10.1111/1462-2920.14626](https://doi.org/10.1111/1462-2920.14626).
19. Talmy D.*, **Beckett S.J.***, Zhang A.B.†, Taniguchi D.A.A., Weitz J.S., Follows M.J. (2019) Contrasting Controls on Microzooplankton Grazing and Viral Infection of Microbial Prey. *Frontiers in Marine Science* 6:182. DOI: [10.3389/fmars.2019.00182](https://doi.org/10.3389/fmars.2019.00182).
20. **Beckett S.J.**, Weitz J.S. (2018) The Effect of Strain Level Diversity on Robust Inference of Virus-Induced Mortality of Phytoplankton. *Frontiers in Microbiology* 9:1850. DOI: [10.3389/fmicb.2018.01850](https://doi.org/10.3389/fmicb.2018.01850).
Associated code: <https://github.com/sjbeckett/DilutionMethod-ViralLysisEstimation>.
21. Weitz J.S., **Beckett S.J.**, Brum J.R., Cael B.B., Dushoff J. (2017) Lysis, lysogeny and virus-microbe ratios. *Nature* 549(7672): E1–E3. DOI: [10.1038/nature23295](https://doi.org/10.1038/nature23295).
Associated code: <https://github.com/WeitzGroup/VMR-Lysis-Lysogeny-v3>.
22. **Beckett S.J.**, Weitz J.S. (2017) Disentangling niche competition from grazing mortality in phytoplankton dilution experiments. *PLOS one* 12(5): e0177517. DOI: [10.1371/journal.pone.0177517](https://doi.org/10.1371/journal.pone.0177517).
Associated code: <https://github.com/sjbeckett/DilutionMethod-NicheCompetition>.
23. **Beckett S.J.** (2016) Improved community detection in weighted bipartite networks. *Royal Society Open Science* 3: 140536. DOI: [10.1098/rsos.140536](https://doi.org/10.1098/rsos.140536).
Associated code: <https://github.com/sjbeckett/weighted-modularity-LPAwbPLUS>.
24. Cowley L.A., **Beckett S.J.**, Chase-Topping M., Perry N., Dallman T.J., Gally D.L., Jenkins C. (2015) Analysis of whole genome sequencing for the *Escherichia coli* O157:H7 typing phages. *BMC Genomics* 16: 271. DOI: [10.1186/s12864-015-1470-z](https://doi.org/10.1186/s12864-015-1470-z).
25. **Beckett S.J.**, Boulton C.A., Williams H.T.P. (2014) FALCON: a software package for analysis of nestedness in bipartite networks. *F1000Research* 3: 185 [v1 ;ref status: indexed, <http://f1000r.es/3z8>]. DOI: [10.12688/f1000research.4831.1](https://doi.org/10.12688/f1000research.4831.1).
Associated code: <https://github.com/sjbeckett/FALCON>.
26. Watts A.J.R., Lewis C., Goodhead R.M., **Beckett S.J.**, Moger J., Tyler C.R., Galloway T.S. (2014) Uptake and retention of microplastics by the shore crab *Carcinus maenas*. *Environmental Science & Technology* 48(15): 8823-8830. DOI: [10.1021/es501090e](https://doi.org/10.1021/es501090e).
27. **Beckett S.J.**, Williams H.T.P. (2013) Coevolutionary diversification creates nested-modular structure in phage-bacteria interaction networks. *Interface Focus* 3: 20130033. DOI: [10.1098/rsfs.2013.0033](https://doi.org/10.1098/rsfs.2013.0033).

Publications in Progress

1. Harris M.J.*, Sinclair A.H.*, Pullano G.*, **Beckett S.J.***, LeJeune L.*, Agosto F.B., Bauch C.T., Baur C., Berestycki H., Dushoff J., Griette Q., Levin S.A., Velasco-Hernández J.X., Wu J., Weitz J.S. Emerging Frontiers in Infectious Disease Modeling: Reassessing the Data-Driven Feedback Loop Between Human Behavior and Disease Dynamics. Accepted at Royal Society Open Science.
 2. Harris M.J., Arani A., Goel T., Zhang K., **Beckett S.J.**, Lo C.N., Dushoff J., Weitz J.S. Interpreting Breakthrough Infections Given Assortative Mixing of Partially Vaccinated Populations. In review. Preprint: <https://doi.org/10.64898/2026.01.22.26344544>. Associated code: <https://github.com/WeitzGroup/Breakthrough-Infections/>.
 3. Sinclair, A.H., Taylor, M.K., **Beckett S.J.**, Chande A.T., Weitz J.S., Samanez-Larkin G.R. Personalized Feedback about Immunity Corrects Risk Misestimation and Motivates Vaccination. In review. Preprint: <https://doi.org/10.31234/osf.io/wgb9t>. Associated code and data: <https://osf.io/74sz9/>.
 4. Stern K., Vincent J., Taranu Z.E., Fortin N., Martel A., Robbe A., Thaize R., Castelli M., **Beckett S.J.**, Posiot T., Buckling A., Shapiro B.J., Tromas N. Deciphering the dynamics of cyanobacteria-phage in a natural lake: insights from a decade-long investigation. In submission. Preprint: <https://doi.org/10.1101/2025.02.19.639072>.
 5. Lindell D., Carlson M.C.G., Weissenbach J., Kirzner S., Šulčius S., Hulata Y., Sabehi G., Sacks J.S., Björkman K.M., Dugenne M., Zborowsky S., Linney M.D., **Beckett S.J.**, Tahan R., Demory D., White A.E., Weitz J.S., Karl D.M., Armbrust E.V., Ingalls A.E., Ribalet F., Caron D.A. Virus-grazer interplay results in enhanced virus production and particle aggregation during infection of *Prochlorococcus*. In prep.
 6. **Beckett S.J.**, Cheung C.†, Weitz J.S. Estimating variant-dependent SARS-CoV-2 fecal shedding to inform wastewater prevalence estimation. In prep.
 7. **Beckett S.J.**, Dominguez-Mirazo, M., Lee, S., Andris C., Weitz J.S. Spread of COVID-19 through Georgia, USA. 2020. Near-term projections and impacts of social distancing via a metapopulation model. Report. Preprint: <https://doi.org/10.1101/2020.05.28.20115642>. Associated code: https://github.com/sjbeckett/MAGEmodel_covid19_GA.
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Software

- 2025 **IDDcourse**: interactive notebooks used in teaching infectious disease dynamics modeling.
Repository: <https://github.com/sjbeckett/IDDcourse>
Language: Python (using marimo) and Julia (using Pluto.jl)
Description: interactive notebooks to help students explore epidemiological modeling.
- 2024 **muEcology_utility**: some utility functions to aid microbial ecology mathematical modeling.
Repository: https://github.com/sjbeckett/muEcology_utility
Language: Python, R
Description: utility functions to aid modeling and unit conversion in microbial ecology.
- 2021 **localcovid19now**: processing and mapping COVID-19 case data at subnational scales.
Repository: <https://github.com/sjbeckett/localcovid19now>
Language: R
Description: Visualize recent COVID-19 case data across the globe (Beckett et al., 2023).

- 2014 **weighted-modularity-LPAwbPLUS**: Improved community detection in weighted bipartite networks
 Repository: <https://github.com/sjbeckett/weighted-modularity-LPAwbPLUS>
 Languages: Julia, MATLAB, Octave, R
 Description: Quantifying modularity in weighted bipartite networks (see Beckett, 2016)
- 2013 **FALCON**: a software package for analysis of nestedness in bipartite networks
 Repository: <https://github.com/sjbeckett/FALCON>
 Languages: MATLAB, Octave, R
 Description: Assessing nestedness in bipartite networks, described in Beckett et al. 2014.
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Presentations

- 2026 **Chesapeake Aquatic Viral Ecology 2026 network meeting**. *IMET, Baltimore, MD.*
Lightning presentation: Estimating variant-dependent SARS-CoV-2 fecal shedding to inform wastewater prevalence estimation.
- 2025 **ASLO 2025 Aquatic Sciences Meeting**. *Charlotte, NC.*
Poster presentation: Towards trait-based aquatic virology: methods and measurements.
- 2025 **MathBio Seminar, Virginia Tech**. *Blacksburg, VA.*
Invited talk: Infections are not alike: models, data, and communication.
- 2025 **Quantitative Ecological and Evolutionary Dynamics seminar**. *UMD, College Park, MD.*
Seminar: Implementing Scientific Code Review: one path towards tackling the reproducibility crisis. (joint w./ Marian Dominguez-Mirazo)
- 2024 **Brin MRC 2024 workshop on Disease Dynamics and Human Behavior**. *UMD, College Park, MD.*
Invited talk: Developing public-facing pandemic tooling to promote prosocial decision making.
- 2024 **Models of Infectious Disease Agent Study Network Annual Meeting: 2023**. *Silver Spring, MD.*
Poster presentation: Infections are not alike: the effects of covariation between individual susceptibility and transmissibility on epidemic dynamics.
- 2024 **Simons Collaboration on Ocean Processes and Ecology Annual Meeting 2024**. *NYC, NY.*
Invited speaker, and poster presentation: Towards generalizations of viral life-history trait scalings.
- 2024 **Pandemic Research Initiative Public Research Seminar**. *University of Maryland, College Park, MD.*
Seminar: Towards a wastewater surveillance and communication nexus. (joint w./ Veerapetch Petchger)
- 2024 **Horn Point seminar, University of Maryland Center for Environmental Science**. *Cambridge, MD.*
Invited talk: Disentangling top-down drivers of mortality in marine microbial communities.
- 2024 **vDARWIN minisymposium**. *Virtual.*
Oral presentation: Quantifying the sensitivity of algal host-virus life-history traits to resource limitation. (joint w/Dr. David Talmy).
- 2024 **Forum for statistically robust dynamic modeling, University of Tennessee, Knoxville**. *Hybrid.*
Invited seminar: Fitting population dynamics to multi-trophic models of *Prochlorococcus*, their viruses, and grazers. (joint w/Dr. David Demory).
- 2024 **Disease ecology and network epidemiology seminar, Georgetown University**. *Washington, DC.*
Seminar: Estimating variant-dependent SARS-CoV-2 fecal shedding to inform wastewater prevalence estimation.
- 2023 **Simons Collaboration on Ocean Processes and Ecology Annual Meeting 2023**. *NYC, NY.*
Invited speaker: Modeling microbial trophic interactions.

- Poster presentation: vDARWIN: Incorporating viral dynamics and ecology in the Darwin Project.*
- 2023 **Models of Infectious Disease Agent Study Network Annual Meeting: 2023.** *Atlanta, GA.*
Lightning talk and poster: Estimating variant-dependent SARS-CoV-2 fecal shedding to inform wastewater prevalence estimation.
- 2023 **Simons Collaboration on Ocean Processes and Ecology Annual Virtual Meeting 2023.** *Virtual.*
*Poster Presentation: Disentangling top-down drivers of mortality underlying diel population dynamics of *Prochlorococcus* in the North Pacific Subtropical Gyre.*
- 2023 **Ocean microbial ecology seminar, University of Tennessee, Knoxville.** *Knoxville, TN.*
Seminar: Modeling virus impacts across systems: from marine microbial communities to COVID-19.
- 2023 **The Serrapilheira/ICTP-SAIR Training Program in Quantitative Ecology.** *São Paulo, Brazil.*
Oral presentation: Applying quantitative principles to develop public facing tools for COVID-19.
- 2021 **Aquatic Viral Workshop 10.** *Virtual/Kyoto (Japan).*
*Oral presentation: Viral lysis, grazing and unaccounted *Prochlorococcus* losses in diel population dynamics in the NPSG.*
- 2021 **Models of Infectious Disease Agent Study Network Annual Meeting: 2021.** *Virtual.*
Poster presentation: Regional risk assessments: bringing actionable information to locales.
- 2020 **Simons Collaboration on Ocean Processes and Ecology Annual Meeting 2020.** *Virtual.*
*Poster presentation: Viral lysis, grazing and unaccounted *Prochlorococcus* losses revealed in diel population dynamics in the North Pacific Subtropical Gyre.*
- 2020 **Oceanography department seminar, Louisiana State University.** *Virtual/Baton Rouge, LA.*
Invited speaker: Modeling virus impacts across systems: from marine microbial communities to COVID-19.
- 2020 **Biology department seminar, University of Illinois at Chicago.** *Chicago, IL.*
Invited speaker: Quantifying the ecological relevance of grazing and viral lysis in marine microbial communities.
- 2020 **Centre for Microbial Dynamics and Infection seminar at Georgia Tech**
Oral presentation: Quantifying the ecological relevance of grazing and viral lysis in marine microbial communities.
- 2020 **Physics of Living Systems seminar at Georgia Tech.**
Oral presentation: Quantifying the ecological relevance of grazing and viral lysis in marine microbial communities.
- 2020 **Ocean Sciences Meeting 2020.** *San Diego, CA.*
*Poster presentation: A day in the life of *Prochlorococcus*: Diel ecological oscillations of cyanobacteria, viruses and grazers in the North Pacific Subtropical Gyre.*
- 2019 **Marine Biological Association seminar.** *Plymouth, UK*
Invited speaker: Computational Marine Microbiology: Linking cellular interactions to population dynamics and ecosystem function.
- 2019 **Simons Collaboration on Ocean Processes and Ecology Annual Meeting 2019.** *NYC, NY.*
*Poster presentation: Estimating *Prochlorococcus* loss rates in north Pacific surface waters associated with viruses, grazers and “other”.*
- 2018 **Simons Collaboration on Ocean Processes and Ecology Annual Meeting 2018.** *NYC, NY.*
Invited speaker and additional poster presentation: Diel with it: Data-model comparisons of diel ecological oscillations around station ALOHA.
- 2018 **5th Postdoctoral Research Symposium at Georgia Tech.**

- Oral presentation: Viral Lysis vs. Grazing: Perspectives on Phytoplankton Mortality.*
- 2018 **Aquatic Viral Workshop 9.** *Lincoln, NE.*
Oral presentation: Viral Lysis vs. Grazing: Perspectives on Phytoplankton Mortality.
- 2018 **Ocean Sciences Meeting.** *Portland, OR.*
Poster presentation: The Effect of Strain Level Diversity on Inference of Grazing and Viral-Induced Mortality.
- 2018 **Suddath Symposium: The Chemical Ecology of Microbiome Interactions.** *Atlanta, GA.*
Poster presentation: Mortality in a bottle: nonlinear feedbacks and biases when inferring viral-induced lysis of plankton.
- 2017 **Marine Biological Association & SAHFOS seminar.** *Plymouth, UK.*
Invited speaker: Quantifying the ecological relevance of viral lysis in complex microbial communities.
- 2017 **Simons Collaboration on Ocean Processes and Ecology Annual Meeting 2017.** *NYC, NY.*
Poster presentation: Mortality in a bottle: nonlinear feedbacks and biases when inferring viral-induced lysis of plankton.
- 2017 **Microbial Dynamics seminar at Georgia Tech.**
Invited speaker: Estimating viral impacts on marine phytoplankton.
- 2017 **ASLO 2017 Aquatic Sciences Meeting.** *Honolulu, HI.*
Oral presentation: Robustness and biases in estimating viral-induced plankton mortality.
- 2016 **Simons Collaboration on Ocean Processes and Ecology Annual Meeting 2016.** *NYC, NY.*
Poster presentation: Theoretical ecology at sea: interpreting mortality rate measurements.
- 2016 **NAKFI conference: Discovering the Deep Blue Sea: Research, Innovation, Social Engagement.** *Irvine, CA.*
Poster presentation: Competition, diversity and disease: implications for plankton mortality rates.
- 2016 **3rd Postdoctoral Research Symposium at Georgia Tech.**
Poster presentation: Towards Modifying the Modified Dilution Method: Robustness and Biases in Estimating Viral-induced Plankton Mortality.
- 2016 **School of Biology retreat.** *Helen, GA.*
Poster presentation: Towards Modifying the Modified Dilution Method: Robustness and Biases in Estimating Viral-induced Plankton Mortality.
- 2016 **Viruses of Microbes 2016.** *Liverpool, UK.*
Poster presentation: Towards Modifying the Modified Dilution Method: Robustness and Biases in Estimating Viral-induced Plankton Mortality.
- 2016 **Aquatic Viral Workshop 8.** *Plymouth, UK*
Oral presentation: Towards Modifying the Modified Dilution Method: Robustness and Biases in Estimating Viral-Induced Plankton Mortality.
- 2015 **Simons Collaboration on Ocean Processes and Ecology Annual Meeting 2015.** *NYC, NY.*
Poster presentation: Diel or no diel? Effectiveness of the dilution method for determining host mortality rates due to viruses.
- 2015 **Living systems: from interaction patterns to critical behavior.** *Venice, Italy.*
Oral presentation: Can coevolution drive phage-bacteria network structure?
- 2014 **Student Conference on Complexity Sciences.** *Brighton, UK.*
Oral presentation: The usage of nestedness for the analysis of bipartite networks.
- 2013 **Disease Group Seminar.** *University of Exeter (Penryn campus), Penryn, UK.*

- Invited speaker:* Coevolved nestedness and modularity in model phage-bacteria infection networks.
- 2013 **Mathematical Models in Ecology and Evolution.** York, UK.
Oral presentation: Coevolved Nestedness and Modularity in model Phage-Bacteria Infection Networks.
- 2013 **Uncertainty in Interaction Networks.** Bath, UK.
Poster presentation: Coevolved nestedness and modularity in phage-bacteria infection networks.
- 2013 **Cambridge Networks Day.** Cambridge, UK.
Poster presentation: Coevolved nestedness and modularity in phage-bacteria infection networks.
- 2013 **Modelling Biological Evolution 2013: Recent Progress, Current Challenges and Future Directions.** Leicester, UK.
Poster presentation: Coevolved nestedness and modularity in phage-bacteria infection networks.
- 2012 **Viruses of Microbes 2012.** Brussels, Belgium.
Poster presentation: Towards trait-based models for aquatic virology.
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Teaching

Lecturing

- 2026 **Introduction to Python programming for Life Sciences** (lead: Stephen Beckett)
University of Maryland
 Fundamental programming skills, and exploring applications within the life sciences. (13 undergraduate students)
- 2025 **Summer School on Quantitative Phage-Bacteria Dynamics Across Scales.**
University of Maryland.
 Lecture (w./ Daniel Muratore and Paul Frémont) on "Modeling phage-bacteria dynamics in oceans" to 30 graduate students.
- 2025 **Quantitative and Computational Biosciences** (course lead: Joshua Weitz)
University of Maryland.
 Guest lecture on: Responding to COVID-19: from models to communication.
- 2025 **Infectious disease dynamics: a systems approach** (co-leads: Stephen Beckett and Gabi Steinbach).
University of Maryland.
 Developed and led course (5 undergraduate + 3 graduate students) focused on using systems thinking to explore infectious disease population dynamics, epidemiological modeling and strategies of disease prevention and mitigation for a range of diseases in human and environmental contexts.
- 2025 **Reading Public Health Emergencies Through Information and Communication Research and Policy** (course lead: Cynthia Baur).
University of Maryland.
 Guest lecture on: Communications from mitigation and surveillance measures.
- 2023 **The Serrapilheira/ICTP-SAIFR Training Program in Quantitative Ecology.**
São Paulo, Brazil.
 Quantitative Foundations of Ecological and Evolutionary concepts. Two week course to 30 Brazilian undergraduate/masters students on concepts and modeling for ecology, evolution and epidemics.

2019, 2018, 2017 **Foundations of Quantitative Biosciences** (course lead: Joshua Weitz).
Georgia Institute of Technology.
Guest lectures on:
* Predator-prey dynamics and evolutionary ecology
* Evolutionary ecology and adaptive dynamics

2018 **Special Topic: Physics of Living Systems** (course lead: Daniel Goldman).
Georgia Institute of Technology.
Guest lectures on:
* Predator-prey dynamics and evolutionary ecology

Workshop Instructor

2021, 2020 Quantitative Biosciences Workshop on Epidemic modeling. *Georgia Institute of Technology.*

Professional development

2023 How does code and science get published?, for The Serrapilheira/ICTP-SAIFR Training Program in Quantitative Ecology. *São Paulo, Brazil.*

2019 Inside the peer review process, for the Quantitative Biosciences Graduate Program. *Georgia Institute of Technology.*

Teaching assistant

2012-2015 Multiple courses including “Introduction to Ecology” and “Modelling for Biosciences”.
University of Exeter.

Mentorship

Undergraduate students

2021-2023 **Caitlin Cheung**, *Undergraduate in Biomedical Engineering, GT*
Data-driven insights to impacts of COVID-19 across county scales in the USA; and estimating temporal evolution of risk with respect to testing; and focus on interpreting wastewater surveillance data. Caitlin is now pursuing an MS in Computational Biology and Quantitative Genetics at Harvard.

2020-2023 **Quan Nguyen**, *Undergraduate in Chemical & Biomolecular Engineering, GT*
Developing interactive dashboards focused on COVID-19 population-level immunity; and COVID-19 risk assessment analysis across multiple countries. Research published in *Nature Human Behaviour & Annals of Epidemiology*. Quan is now pursuing a PhD at U.Penn.

2019-2020 **Robert Morgan**, *Undergraduate in Biological Sciences, GT*
Visualization of and interactive tutorials for ecological models. Supported Robert in applying for a President’s Undergraduate Research Awards at Georgia Tech.

2017-2018 **Adam Zhang**, *Undergraduate in Mathematics, GT*
Research contributed to the manuscript: “Contrasting controls on microzooplankton grazing and viral infection of microbial prey”. Adam completed a Data Science Internship at UPS and is now an Applied AI manager at Optimal Dynamics.

Graduate students

2025-Present **Kejia Zhang**, *PhD student in Applied Mathematics & Statistics, and Scientific Computation, UMD*

Exploring the effects of immunity waning and susceptibility heterogeneity in epidemiological models.

2024-Present **Akash Arani**, *PhD student in Biology, UMD*

Population-level models incorporating the effects of behavioral feedback on vaccination, as well as waning immunity and reinfection. Akash attended SFI's Complex Systems Summer School in 2025.

2022-Present **Raunak Dey**, *PhD student in Physics, GT then UMD*

Model-data integration and inference in ecological communities. Raunak has been awarded the UMD Microbiome Center Summer Scholarship, Thomas G. Mason Interdisciplinary Physics Fund, and International Conference Student Support Award.

2018-2019 **GuanLin Li**, *PhD in Quantitative Biosciences, GT*

Timeseries inference and evolutionary dynamics of virus-microbe systems. GuanLin is now a Quantitative Researcher at China Securities Co., Ltd.

2017-2019 **Daniel Muratore**, *PhD in Quantitative Biosciences, GT*

Microbial oceanographic modelling, game-theory and time-series data analysis. Daniel is now a Complexity Postdoctoral and Omidyar Fellow at the Sante Fe Institute.

2016-2021 **Ashley Coenen**, *PhD in Physics, GT*

Inferring ecological virus-host community structures from population dynamics. Ashley is now a Business Analyst and Data Scientist at NRG Energy.

2016-2017 **Charles Wigington**, *PhD in Bioinformatics, GT*

Quantifying relationships of oceanic virus-to-microbe ratios. Charles is now a Data Scientist at Insulet Corporation.

2015-2018 **Shengyun Peng**, *PhD in Bioinformatics, GT*

Investigating virus-host infection from single cell and genomic perspectives. Shengyun is now a Data Scientist with Adobe.

Professional Activities and Service

Conference Organization

Organizer for the *Chesapeake Aquatic Viral Ecology 2026 network meeting*. Co-leads: Eric Schott (IMET), Julie Pourtois (UMD) and Tapan Goel (UMD). Institute of Marine and Environmental Technology, Baltimore, MD, USA (2026). Acquired \$1250 funding via Maryland Sea Grant (PI: Goel).

Organizer for the inaugural *Maryland Aquatic Viral Ecology meeting*. Co-leads: Julie Pourtois (UMD) and Tapan Goel (UMD). University of Maryland, College Park, MD, USA (2025).

Co-chair at the *ASLO 2025 Aquatic Sciences Meeting* for the session: *Viral interactions and ecological dynamics*. Co-leads: Elaine Luo (UNCC) and Kristina Mojica (USM). Charlotte, NC, USA (2025).

Co-chair at *2018 Ocean Sciences Meeting* for the session: *A Matter of Life and Death: The Role of Microbial Interactions in Mediating Biogeochemical Cycles*. Co-lead with David Talmy (UTK), Kyle Mayers (SOTON) and Elizabeth Harvey (UGA). Portland, OR, USA (2018).

Organizer for the *4th Postdoctoral Research Symposium at Georgia Tech*. Acquired \$8,500 in funds for awards to outstanding presenters, organized sessions, reviewed abstracts and coordinated the symposium. I also convened the symposium. Atlanta, GA, USA (2017).

Memberships

Association for the Sciences of Limnology and Oceanography (ASLO); Models of Infectious Disease Agent Study (MIDAS); International Society for Viruses of Microorganisms (ISVM), Chesapeake Aquatic Viral Ecology (CAVE) network, American Society for Microbiology (ASM), Global Society for Infectious Disease Dynamics (GSIDD).

Editorial Service:

Associate Editor, Mathematics in Medicine and Life Sciences, (Oct 2024 – present)

Academic Editor, PLOS Computational Biology (Jan 2025 – present)

Special Issue Editor: Behavioral Epidemiology, for Mathematics in Medicine and Life Sciences (2024-2025).

Guest Editor for PNAS (2026).

Scientific Peer Review

Reviewer for: Applied Network Science, Axios Review, BMJ Open, Communications Biology, Concurrency & Computation: Practice & Experience, Current Opinion in Systems Biology, Diversity, Ecology Letters, eLife, Europhysics Letters, Evolutionary Bioinformatics, FEMS Microbiology Ecology, Food Webs, Frontiers in Ecology and Evolution, Frontiers in Genetics, Frontiers in Marine Science, IEEE Access, IEEE Transactions on Computational Social Systems, Journal of Biological Physics, Journal of Intelligent & Fuzzy Systems, Journal of The Royal Society Interface, Knowledge and Information Systems, Limnology and Oceanography, mBio, Methods in Ecology and Evolution, Molecular Ecology, Nature Communications, Network Science, Oikos, Physica A: Statistical Mechanics and its Applications, PLOS Complex Systems, PLOS Computational Biology, PLOS ONE, Scientific Reports, The American Naturalist, The ISME Journal, Trends in Microbiology, Viruses.

Laboratory Experience

- *Mortality Workshop*. Technion - Israel Institute of Technology, Haifa, Israel (Sept., 2019).

Part of a team investigating mortality of *Prochlorococcus* by viruses and grazers *in vitro* (hosted by lab of Prof. Debbie Lindell).

Outreach

- Founding organizer for the Chesapeake Aquatic Viral Ecology network.
- [Ask Me Anything](#), via reddit/AskScience (>26M members) on epidemic modeling and pandemic prevention – jointly with Mallory Harris and Joshua Weitz (2024).
- Interviewed with two high school students in Montgomery county, Maryland about my work and the scientific process (2023).
- Collaborative work contributing to the [COVID-19 Immunity Estimator](#) tool, to help inform individuals about waning immunity and their risk of exposure and severe disease (2023).
- Collaborative work contributing to [COVID-19 population-level immunity dashboards](#) to show how natural infections and vaccinations contribute to herd immunity (2021-2022).
- Collaboration [with Mary Wang](#) (Georgia Tech.) and Science.Art.Wonder to promote science through art. December 2020-March 2021. Science.Art.Wonder Showcase. Artwork: “Us”.
- Collaborative work contributing to [interactive COVID-19 event risk map dashboards](#) for mapping risk in regional areas across multiple countries.
- Interactive data dashboard of [COVID-19 metrics in Georgia](#) as an interactive Rshiny app.
- Data Visualisations of [COVID-19 spread in Georgia](#), showing time-lapses for recorded cases and deaths. April 2020.
- Hosted the science communication Twitter account @biotweeps (audience of over 18,000) for a week in [March 2019](#), where I talked about modelling and marine viruses.

- Collaboration [with Emily Madsen](#) (Georgia Tech.) and Science.Art.Wonder to promote science via art. December 2018-April 2019. Atlanta Science Festival and Clough Art Crawl. Artwork: “Submerged” and “Through a Different Model”.
- Collaboration with ceramic sculptor RJ Sturgess (Georgia State) and Science.Art.Wonder to promote science through art at the Atlanta Science Festival. December 2017 - March 2018. Artwork: “Delicate Balance” and “Phage studies 1-3”.
- Educator supporting a massive open online course (MOOC) run by the University of Exeter and FutureLearn titled “*Climate Change: Challenges and Solutions*”. Engaged with learners online and via weekly feedback videos. 2014.
- Run “[Phage on toast](#)” where I occasionally blog about my research and research experiences. Over 1,750 visitors from 73 countries. (2013 – present).

Media

- **Beckett S.J., Weitz J.S.** 2020. [Georgia’s Reopening Depended on Missing Data](#). *Slate* (Medical Examiner column). May 2020.

Press

- **UMD CMNS:** [Biology Professor Joshua Weitz Answers Questions About Pandemic Prevention](#)
- **Courier Post:** [Holiday shopping in-person this year? Here are some ways to lower COVID risk](#)
- **RStudio:** [Communicating with 8 Million People through Shiny](#)
- **Montana Public Radio:** [How Risky Are Holiday Gatherings? Here's What Health Experts Say About COVID-19](#)
- **WOGX Orlando:** [Interactive map shows your risk of catching COVID-19 at an event](#)
- **WABAY-TV Green Bay:** [Interactive tool helps you understand risk factor of a gathering](#)
- **WDAF-TV Kansas City:** [Thanksgiving plans this year may be changing due to COVID-19](#)
- **WHBQ-TV Memphis:** [Researchers develop risk calculation tool for COVID-19 exposure](#)
- **KSNV News 3 Las Vegas:** [Tool shows risk of catching COVID-19 in group settings](#)
- **KADN News15, Lafayette & Acadiana:** [COVID-19 Risk Assessment Map](#)
- **MetroLab/Government Technology:** [Mapping Tool Visualizes COVID Risk of Different-Size Events](#). October 2020 Innovation of the month.
- **Business Insider:** [A 'cuckoo' graph with no sense of time or place shows how Georgia bungled coronavirus data as it reopens](#)
- **WABE:** [Social Distancing Reduced Virus Spread By 50% In Georgia, Study Finds](#)
- **GaTech College of Sciences:** [Science Inspiring Art](#)

Other Service

- Reviewer for *Models of Infectious Disease Agent Study Network Annual Meeting: 2025*. Bethesda, MD, USA (2025).
- Reviewer for *Models of Infectious Disease Agent Study Network Annual Meeting: 2024*. Silver Spring, MD, USA (2024).
- Member of the [Pandemic Readiness Initiative](#) at UMD (2023-).
- Reviewer for *Models of Infectious Disease Agent Study Network Annual Meeting: 2023*. Atlanta, USA (2023).
- Judge at the *15th Annual Undergraduate Research Spring Symposium*, Georgia Tech. Atlanta, USA (2021).
- Reviewer for President's Undergraduate Research Awards at Georgia Tech. Atlanta (2019).
- Judged at the *12th Annual Undergraduate Research Spring Symposium*, Georgia Tech. Atlanta, USA (2017).

- Lead GitHub demonstrations within the Weitz group at Georgia Tech to improve and aid the production of reproducible computational science (2015, 2016).
 - Assisted with the Exeter Climate Change Exhibit at *Transformational Climate Science* conference. Exeter, UK (2014).
 - Session chair at *Mathematical Models in Ecology and Evolution*. York, UK (2013).
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